



CS355 - Digital Forensics

Report for Labs 2-4

Archie Harrodine
ID:2127935

February 13, 2024

Contents

1	Lab 2	3
	1.1 Exercise 1	3
	1.2 Exercise 2	3
2	Lab 3	3
	2.1 Exercise 1	3
3	Lab 4	4
	3.1 Exercise 1	4
	3.2 Exercise 2	4

1. Lab 2

1.1. Exercise 1

- (c) 6 figures (R, G, B and Y', Cb, Cr images) for Exercise 1
These are the RGB images¹, and these are the Y'CbCr images².

1.2. Exercise 2

- (d) 3 figures (Y', Cb, Cr images) and their dimensions for Q1 in Exercise 2
These are the Y'CbCr images³. The dimensions of Y' are 1080×1920 , and the dimensions of Cb and Cr are 540×960 (in the format of image height $\times$ image width).
- (e) the output suspectID from (b)
The output of suspectID is 'D', with the number of blocks with the highest similarity being 1578.

2. Lab 3

2.1. Exercise 1

- (b) Submit a brief report, and answer the following questions. 1). How many parameters did you tune? 2). Briefly explain your rationale behind your parameter choices to produce the clean image. 3). Display the clean image.

The method used to remove notch noise was to first visualise a shifted centred frequency image, which let me see the points of greatest frequency. Since noise is mostly centred around points of high frequency, and our noise was periodic, I identified the points of repeating high frequency, to which multiple filters would be applied together in that area to work in tandem to reduce noise while not greatly reducing image quality. Working in increments of 256 (going from $[-512, -512]$ up to $[512, 512]$), I applied a lowpass ideal, Gaussian and Butterworth filter of each the same parameters. The point 0,0 was excluded as that high frequency represented key information from the image.

The parameters tuned were $D0$ in all three filters, and n in the Butterworth filter. So technically we tuned 4 different parameters, but you could argue its 2? Point is, these are the parameters that were changed around. The final values decided upon was for ideal: $D0 = 30$, Gaussian: $D0 = 60$ and Butterworth $D0 = 220, n = 0.9$. For ideal and Gaussian filters, $D0$ acted as the effective radius that the filter would affect in the frequency visual. Ideal would completely get rid of whatever it was at, while Gaussian would gently remove the rest. By removing the periodic high frequency points, the ideal filter radius was tuned to get entirely rid of the strongest part; while then the Gaussian filter would then gently remove other high frequencies around the high points, but not too much as to remove the frequencies of the undistorted image. Hence the ideal filter is about half the radius($D0$) of the Gaussian filter. The Butterworth filter is slightly different, in which $D0$ does act like a radius, but its effects

are far weaker and more sparse, hence the much larger radius. This makes the image far less grainy and a bit clearer. We can also change n , however increasing or decreasing it significantly past the default of 1.0 did not improve image quality.

The final image quality is fairly strong, and we can make out the driving licence to be 4DSU496. We could make the image more clear if we added a few more smaller filters around the higher frequency lines between the high frequency points. Here is the cleaned image⁴.

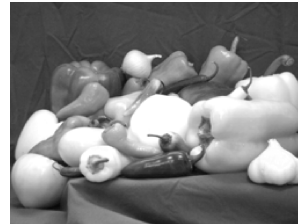
3. Lab 4

3.1. Exercise 1

- (a) Exercise 1(a) display the original Warwick Logo and the binarized Warwick Logo. These are the wanted images⁵.
- (b) Exercise 1(b-c) display the original Baboon image and the watermarked Baboon images: 1) invisible one with LSB substitution 2) visible one with MSB substitution. These are the wanted images⁶.
- (c) Exercise 1(c) SSIM scores between watermarked images and the original Baboon image: 1) SSIM between invisible one and original; 1) SSIM between invisible one and original. SSIM between invisible and original: 0.9994.
SSIM between visible and original: 0.1874.

3.2. Exercise 2

- (d) Exercise 2 (c) display the four extracted/recovered watermarks corresponding to four compression cases, i.e., no compression, compression with quality 100/98/95) These are the wanted images⁷.

RGB Image**Red color****Green color****Blue color****Fig 1:** RGB Images in grayscale for exercise 1.

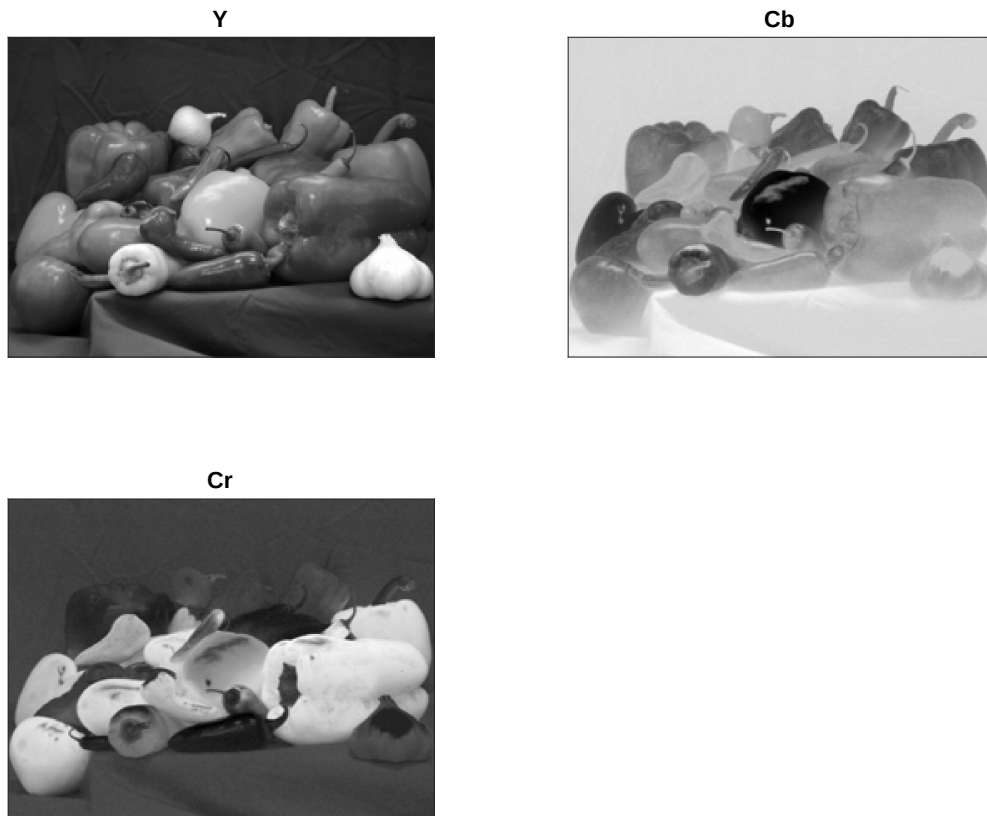


Fig 2: Y'CbCr Images in grayscale for exercise 1.

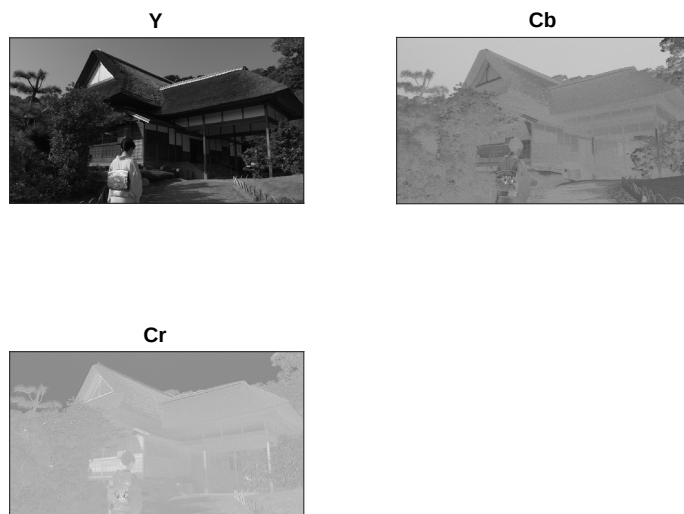


Fig 3: Y'CbCr Images in grayscale for exercise 2.



Fig 4: The cleaned halftone evidence image.



Fig 5: The original logo and the binarized logo.

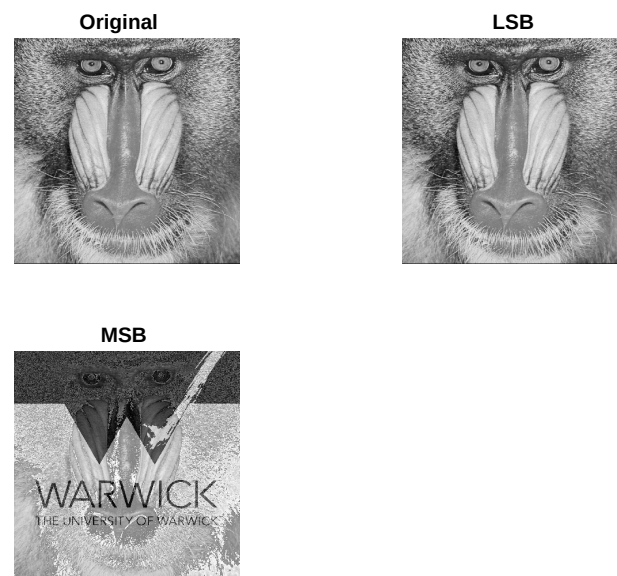


Fig 6: The original image and the MSB & LSB watermarked images.

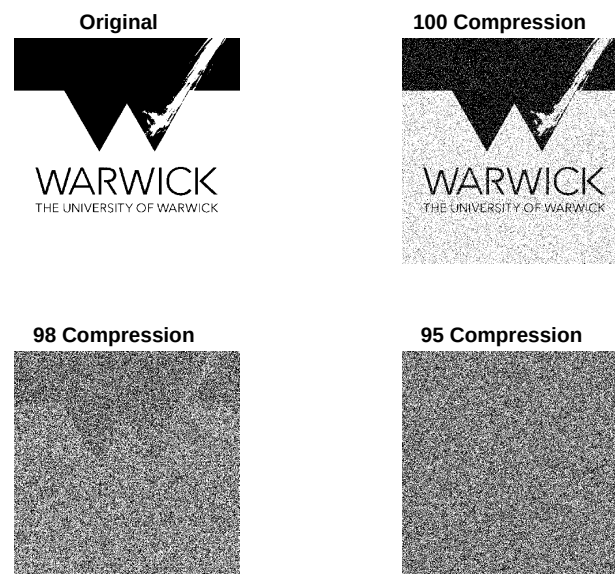


Fig 7: The four recovered watermarks.